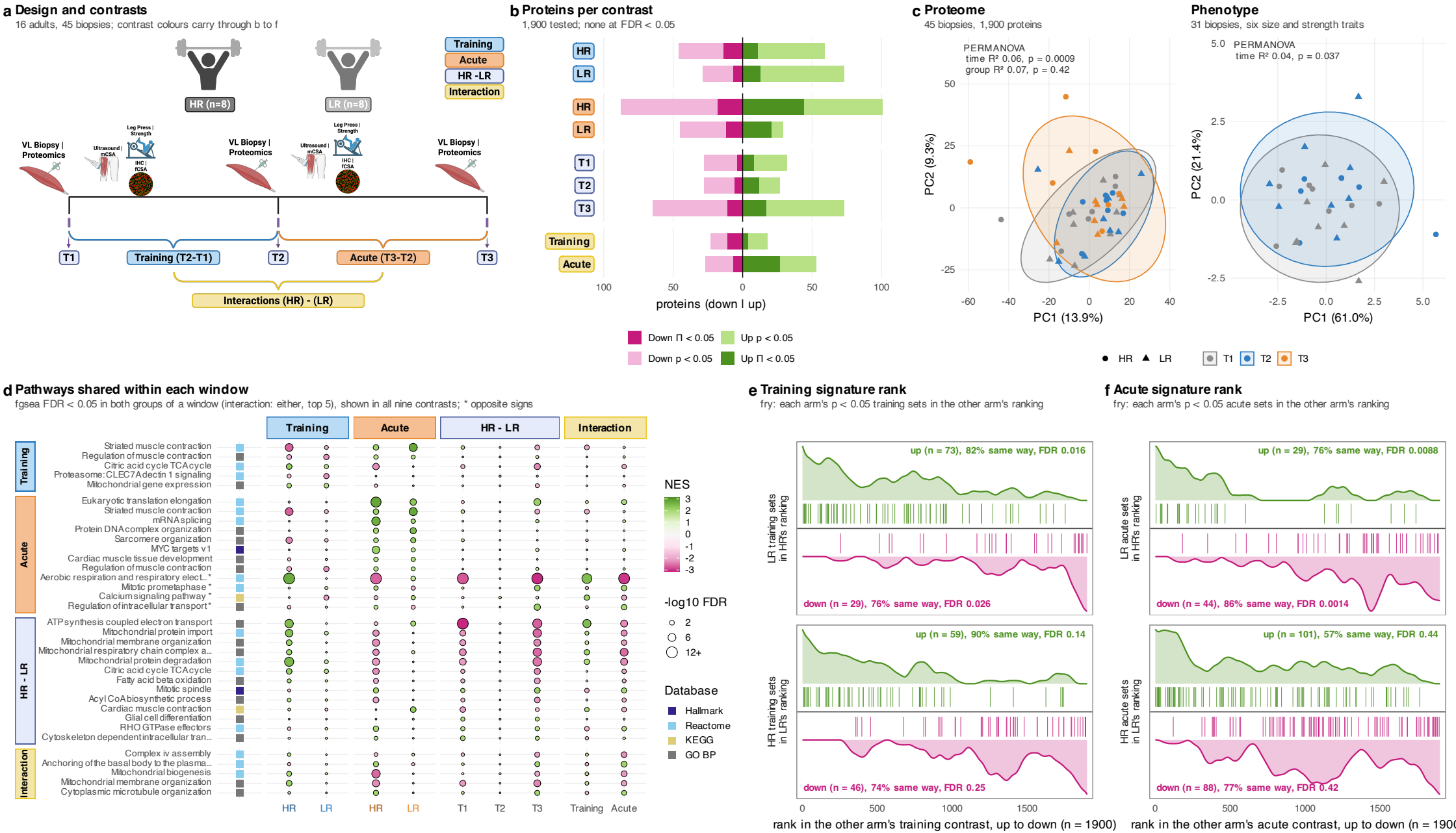

a Design and contrasts
16 adults, 45 biopsies; contrast colours carry through b to f
HR (n=8)
LR (n=8)
Training
Acute
HR -LR
Interaction
VL Biopsy | Proteomics
Ultrasound | mCSA
Leg Press | Strength
IHC | fCSA
T1
Training (T2-T1)
T2
Acute (T3-T2)
T3
Interactions (HR) - (LR)
b Proteins per contrast
1,900 tested; none at FDR < 0.05
HR
LR
T1
T2
T3
Training
Acute
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
45 biopsies, 1,900 proteins
PERMANOVA
time R² 0.06, p = 0.0009
group R² 0.07, p = 0.42
PC1 (13.9%)
PC2 (9.3%)
Phenotype
31 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC1 (61.0%)
PC2 (21.4%)
HR
LR
T1
T2
T3
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Striated muscle contraction
Regulation of muscle contraction
Citric acid cycle TCA cycle
Proteasome: CLEC7A dectin 1 signaling
Mitochondrial gene expression
Eukaryotic translation elongation
mRNA splicing
Protein DNA complex organization
Sarcomere organization
MYC targets v1
Cardiac muscle tissue development
Aerobic respiration and respiratory elect.. *
Mitotic prometaphase *
Calcium signaling pathway *
Regulation of intracellular transport *
ATP synthesis coupled electron transport
Mitochondrial protein import
Mitochondrial membrane organization
Mitochondrial respiratory chain complex a...
Mitochondrial protein degradation
Fatty acid beta oxidation
Mitotic spindle
Acyl CoA biosynthetic process
Glial cell differentiation
RHO GTPase effectors
Cytoskeleton dependent intracellular tran...
Complex iv assembly
Anchoring of the basal body to the plasma...
Mitochondrial biogenesis
NES
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each arm's p < 0.05 training sets in the other arm's ranking
up (n = 73), 82% same way, FDR 0.016
down (n = 29), 76% same way, FDR 0.026
up (n = 59), 90% same way, FDR 0.14
down (n = 46), 74% same way, FDR 0.25
LR training sets in HR's ranking
HR training sets in LR's ranking
rank in the other arm's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each arm's p < 0.05 acute sets in the other arm's ranking
up (n = 29), 76% same way, FDR 0.0088
down (n = 44), 86% same way, FDR 0.0014
up (n = 101), 57% same way, FDR 0.44
down (n = 88), 77% same way, FDR 0.42
LR acute sets in HR's ranking
HR acute sets in LR's ranking
rank in the other arm's acute contrast, up to down (n = 1900)